Mohamed Ashraf

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LinkedIn | GitHub | Portfolio | Streamlit | Gmail

Professional Summary

Results-driven Machine Learning Engineer and Computer Science student with demonstrated expertise in designing and implementing scalable machine learning systems, signal processing algorithms, and database architectures. Proven ability to translate complex technical challenges into production-ready solutions with measurable impact. Specializing in classification algorithms, regression modeling, statistical signal processing, and system-level optimization. Strong problem solver with hands-on experience building end-to-end ML pipelines from data preprocessing to model deployment. Seeking machine learning opportunities in Egypt and the MENA region to advance AI innovation and drive business impact.

Technical Skills

- **Machine Learning & AI:** Supervised Learning (Naive Bayes, Decision Trees, Ensemble Methods, Regression Models), Statistical Modeling, Classification & Regression Algorithms, Hyperparameter Tuning, Model Evaluation & Validation, Feature Engineering, Predictive Analytics
- **Signal Processing:** Analog Communications, Digital Signal Processing (DSP), Modulation Theory, Statistical Signal Analysis, Time-Frequency Analysis
- **Programming Languages:** Python, Java, MATLAB, SQL, C (Systems Programming), M (Power Query)
- **Database & Data Management:** SQL Server, Query Optimization, Database Design, Data Normalization, Transaction Management, Concurrency Control, Data Analysis & Visualization
- **Development Tools & Platforms:** Git/GitHub, Linux/Ubuntu, MATLAB, Streamlit, Jupyter Notebooks, Power BI/Power Query, Command-Line Interface (CLI)
- **Operating Systems & Systems:** Process Management, Memory Management, Concurrent Programming, System Calls, File I/O Operations

Academic Projects & Experience

California House Price Prediction — Machine Learning Project | 2024

- Developed regression model to predict California housing prices using multivariate linear regression and feature scaling techniques
- Implemented comprehensive exploratory data analysis (EDA) identifying key price drivers and feature correlations

- Engineered feature preprocessing pipeline with normalization and outlier handling
- Achieved competitive model performance with R^2 score evaluation and residual analysis
- Built visualization dashboard for model insights and prediction analysis

Repository: github.com/MohamedAshraf-DE/California-House-Price-Prediction

Movie Revenue Prediction System — Applied ML Project | 2024

- Engineered machine learning system predicting movie box office revenue from production features and metadata
- Designed data collection and feature engineering pipeline extracting predictive indicators from film attributes
- Implemented ensemble methods combining multiple regression algorithms for robust predictions
- Developed performance evaluation framework with cross-validation and error analysis
- Created interpretable model outputs for stakeholder communication and decision-making support

Repository: github.com/MohamedAshraf-DE/Movie-Revenue-Prediction

Uber Ride Cancellation Prediction — Predictive Analytics Project | 2024

- Built classification model predicting ride cancellation probability using historical ride data and contextual features
- Implemented class imbalance handling techniques to address skewed cancellation distribution
- Engineered feature set including temporal patterns, driver characteristics, and demand indicators
- Developed model interpretability analysis identifying primary cancellation risk factors
- Optimized model hyperparameters achieving strong precision and recall balance for business applicability
- Deployed interactive Streamlit dashboard with production-ready deployment instructions

Repository: github.com/MohamedAshraf-DE/Uber-Ride-Cancellation-Prediction

DT-Naive-ADA: Multi-Algorithm Classification System — Personal ML Project | Dec 2024 – Present

- Implemented ensemble machine learning pipeline combining Decision Trees, Naive Bayes, and Adaptive Boosting (AdaBoost)
- Developed comprehensive data preprocessing module with automatic feature normalization and missing value handling
- Built interactive Streamlit web application enabling users to upload datasets and visualize model performance metrics

- Engineered model persistence system for saving trained models and enabling production deployment
- Achieved clean code architecture with modular components for scalability and maintainability

Repository: github.com/MohamedAshraf-DE/DT-Naive-ADA

Gamma-Ray vs Hadron Classification System — Applied ML Research Project | Nov 2024 – Dec 2024

- Designed machine learning framework for astrophysics applications to classify gamma-ray and hadron events
- Developed user-friendly interface targeted at non-programmer scientists with interpretable model explanations
- Engineered data upload, visualization, and exploratory data analysis (EDA) capabilities
- Implemented model prediction interface with confidence scores and feature importance visualization
- Created automated report generation system for scientific documentation
- Deployed Production Application

Repository: github.com/MohamedAshraf-DE/Gamma-Hadron

Database Management & Optimization — Coursework | 2024

- Designed normalized database schemas applying entity-relationship modeling and normalization principles
- Wrote complex SQL queries (JOINs, subqueries, aggregations) for efficient data retrieval and analysis
- Implemented transaction management, concurrency control, and ACID compliance
- Optimized query performance through indexing strategies and execution plan analysis

Signal Processing & Modulation Analysis — Coursework | 2024

- Implemented DSP algorithms for analog and digital signal analysis in MATLAB
- Analyzed modulation schemes and their performance under noise conditions
- Conducted statistical signal processing experiments with probability distributions and entropy calculations
- Developed visualization tools for time-frequency analysis and signal characteristics

Technical Certifications & Continuous Learning

Completed Certifications:

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI (Issued: November 2025)
- Getting Started with Deep Learning – NVIDIA (Issued: January 2025)
- IT Trainee – Abou Qir Petroleum (WEPCO) | Introductory training in IT/technology operations

In Progress / Expected Completion:

- Power BI Specialist – DEPI (Expected: January 2026) | Advanced data visualization, business intelligence, and analytics dashboard development
- Machine Learning for Data Analysis – NTI (Expected: January 2026) | Practical machine learning applications for real-world data analysis scenarios

Volunteering & Activities

- **حفل تفوق جامعات مصر (Egyptian Universities Excellence Ceremony):** Recognized for academic excellence and participation in university excellence events. Enhanced communication and networking skills through recognition ceremonies.
- **Arabic Font Competition:** Participated in competitive Arabic typography design competition. Strengthened design sense and attention to detail through creative challenges.
- **Topper Students Competition:** Honored among top-performing students at school-level competition. Demonstrated leadership and academic mastery among peer competitors.
- **Bisco Misr Factory Visit:** Explored large-scale food manufacturing processes, safety standards, and industrial workflows. Developed practical knowledge of real-world business operations.
- **Bibliotheca Alexandrina Programs:** Attended technical and communication skills workshops. Enhanced soft skills: communication, collaboration, leadership, curiosity, and problem-solving.

Freelance & Professional Profiles

Available for freelance projects on multiple platforms:

- Upwork
- Mostaql
- Khamsat
- Freelancer
- Outlier AI

Language Proficiency

- Arabic: Native
- English: Professional Working Proficiency

- German: Elementary/Conversational

Education

Bachelor of Science in Computer and Communication Engineering (In Progress)
Alexandria University, Egypt | GPA: 3.81/4.0
Specialization: Machine Learning, Signal Processing, Database Systems, Operating Systems

Relevant Coursework: Machine Learning Algorithms, Advanced Database Systems, Signal Processing, Operating Systems, Data Structures, Algorithm Analysis, Probability & Statistics, Information Theory, Communications Engineering

Projects Portfolio

- California House Pricing — Regression model with comprehensive feature analysis
- Movie Revenue Prediction — Ensemble regression system for entertainment analytics
- Uber Ride Cancellation — Classification model with Streamlit deployment
- DT-Naive-ADA Classifier — Multi-algorithm ensemble system with Streamlit interface
- Gamma-Hadron Classifier — ML application for astrophysics research
- Streamlit Apps: share.streamlit.io/user/mohamedashraf-de
- Portfolio Website: mohamed-ashraf-github-io.vercel.app
- GitHub Portfolio: github.com/MohamedAshraf-DE

Career Objectives

Seeking a Machine Learning Engineer or Data Scientist position in Egypt and the MENA region to leverage strong academic foundation (GPA 3.81) and practical project experience in building scalable ML systems, contributing to cutting-edge AI initiatives, and advancing career growth in the Egyptian and regional tech ecosystem.

References and detailed project demonstrations available upon request.

Full portfolio and project repositories: https://github.com/MohamedAshraf-DE